

Warehouse Manager

Full-stack, multi-warehouse inventory & logistics platform

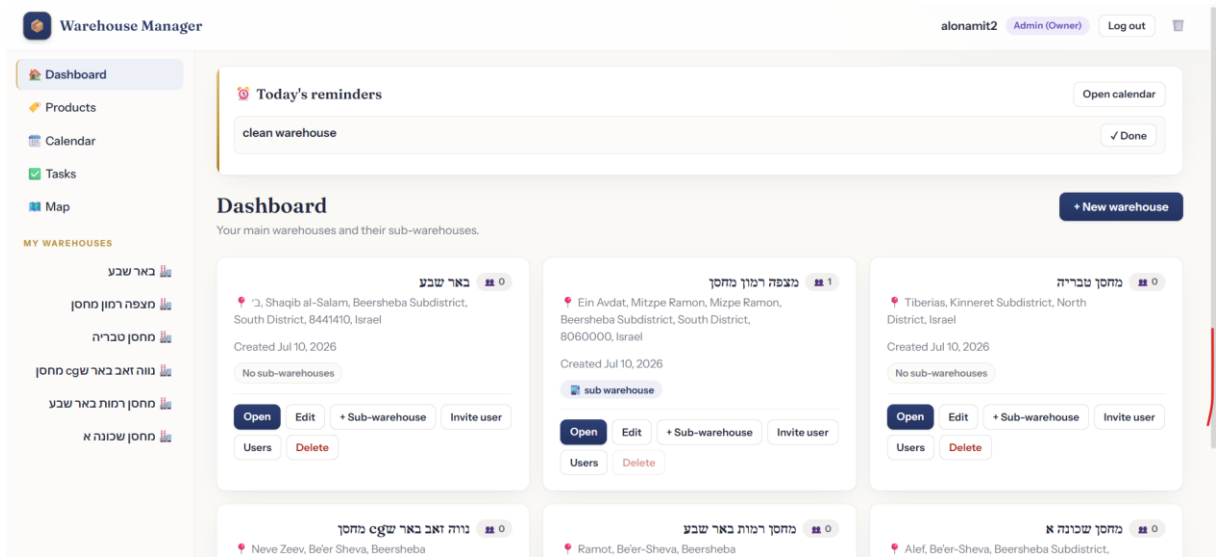
Angular 22 · .NET 10 · SQL Server · JWT + refresh · 72 automated tests

Overview

Warehouse Manager lets a business owner run an entire network of warehouses from one place. Each account owns a tree of main warehouses and nested sub-warehouses, stocks them from a shared product catalog, manages the team that works them, and plans delivery routes between locations. All data is strictly scoped per owner — nothing leaks between accounts.

Core capabilities

- **Warehouse hierarchy** — main warehouses with nested sub-warehouses, each with its own address, team and stock.
- **Shared product catalog** — products defined once (SKU, price, barcode) and placed into any warehouse as stock items.
- **Live inventory** — quantity, minimum level and an auto-computed status (OK / Low / Out of stock).
- **Signed stock movements** — Stock-In / Stock-Out / Sale / Return / Adjustment, with a history that survives item deletion.
- **Roles & team** — Owner, Shift-Manager and Employee, invited per warehouse and enforced on the server.
- **Work schedule** — assign shifts (day + hours + notes); each member sees their own upcoming shifts.
- **Reminders & tasks** — private calendar reminders (today's surface on the dashboard) and a personal to-do list with due dates.
- **Smart route optimizer** — weight- and urgency-aware delivery routing between warehouses (see flagship feature).



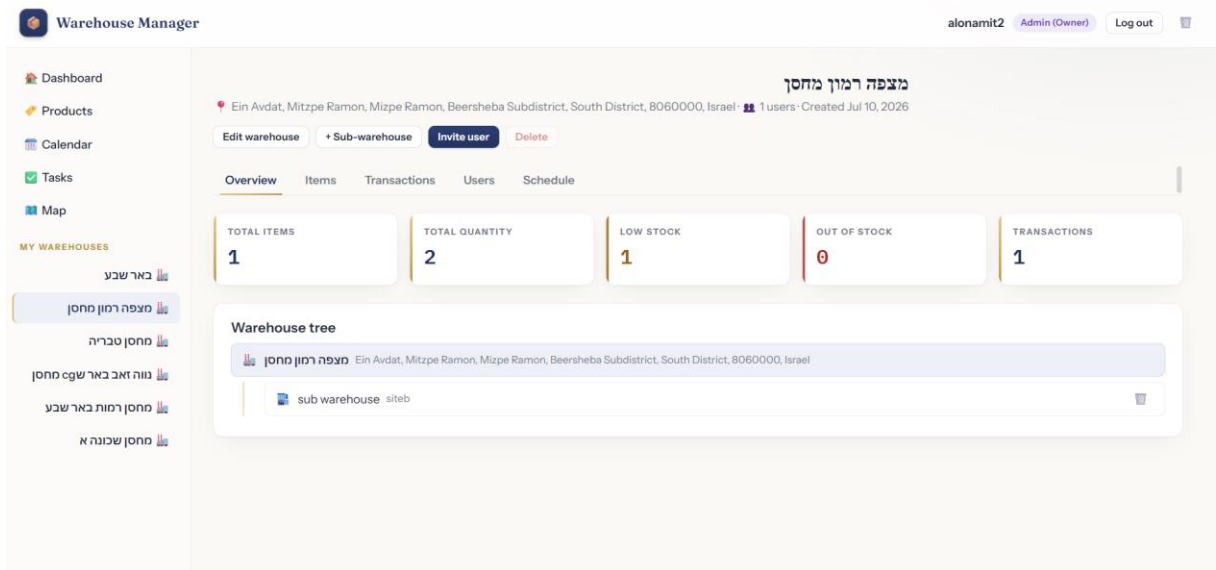
Owner dashboard — every warehouse with its sub-warehouses, team size and quick actions.

The warehouse workspace

Opening any warehouse shows a live snapshot that rolls up its whole subtree, across five tabs: Overview, Items, Transactions, Users and Schedule.

Example — “מחסן רמון מצפה” (Mitzpe Ramon):

Total items	Total quantity	Low stock	Out of stock	Transactions
1	2	1	0	1



Warehouse workspace — rollup stats and the warehouse tree, across five tabs.

Catalog & stock

Products live once in a per-owner catalog; items place a product into a warehouse with a quantity and a minimum level, and the status computes itself.

SKU	Product	Location	Qty	Min	Status
F67	Baseball	shelf B	2	2	Low
A12	Pallet wrap	shelf A	48	20	OK
B03	Label roll	bay 2	0	10	Out

Catalog example: Baseball · \$0.09 · barcode 12121212.

Warehouse Manager

aloramit2Admin (Owner)Log out

Dashboard

Products

Calendar

Tasks

Map

MY WAREHOUSES

באר שבע

מצפה רמון מחסן

מחסן סברה

נווה זאב באר שבע מחסן

מחסן רמות באר שבע

מחסן שכונה א

מצפה רמון מחסן

Ein Avdat, Mitzpe Ramon, Mitzpe Ramon, Beersheba Subdistrict, South District, 8060000, Israel · 1 users · Created Jul 10, 2026

Edit warehouse

+ Sub-warehouse

Invite user

Delete

Overview

Items

Transactions

Users

Schedule

Items (including sub-warehouses)

+ New item

SKU	PRODUCT	LOCATION	WAREHOUSE NODE	QTY	MIN LEVEL	STATUS	
F67	Baseball	shelf B	sub warehouse	2	2	Low	EditDelete

Items table — stock status computed per item; sub-warehouse items roll up.

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Product catalog

Products shared across your warehouses.

+ New product

Search by SKU, name or barco

SKU	NAME	PRICE	BARCODE	CREATED	
F67	Baseball	\$0.09	12121212	Jul 11, 2026	EditDelete

Product catalog — SKU, price and barcode, shared across warehouses.

Team: roles & shifts

- Invite users straight into a warehouse and assign **Employee** or **Shift-Manager** roles (Admin invitations are rejected).
- Remove members; owners keep full control of their tree.
- Assign shifts — e.g. **alon2 • Tue, Jul 14 • 08:00–16:00**. Employees see their own upcoming shifts on their warehouse page.

Warehouse Manager

alonamit2 Admin (Owner) Log out

מבצפה רמון מחסן

Ein Avdat, Mitzpeh Ramon, Mizpeh Ramon, Beersheba Subdistrict, South District, 8060000, Israel • 1 users • Created Jul 10, 2026

Edit warehouse + Sub-warehouse Invite user Delete

Overview Items Transactions Users Schedule

Users

USERNAME	EMAIL	ROLE	WAREHOUSE
alon2	amitu4444@gmail.com	Employee	מבצפה רמון מחסן

Log out

Users tab — invite team members and assign roles.

Warehouse Manager

alonamit2 Admin (Owner) Log out

מבצפה רמון מחסן

Ein Avdat, Mitzpeh Ramon, Mizpeh Ramon, Beersheba Subdistrict, South District, 8060000, Israel • 1 users • Created Jul 10, 2026

Edit warehouse + Sub-warehouse Invite user Delete

Overview Items Transactions Users Schedule

Assign a shift

Team member Date Start End

alon2 14/07/2026 08:00 16:00

Notes (optional)

+ Assign shift

Upcoming schedule

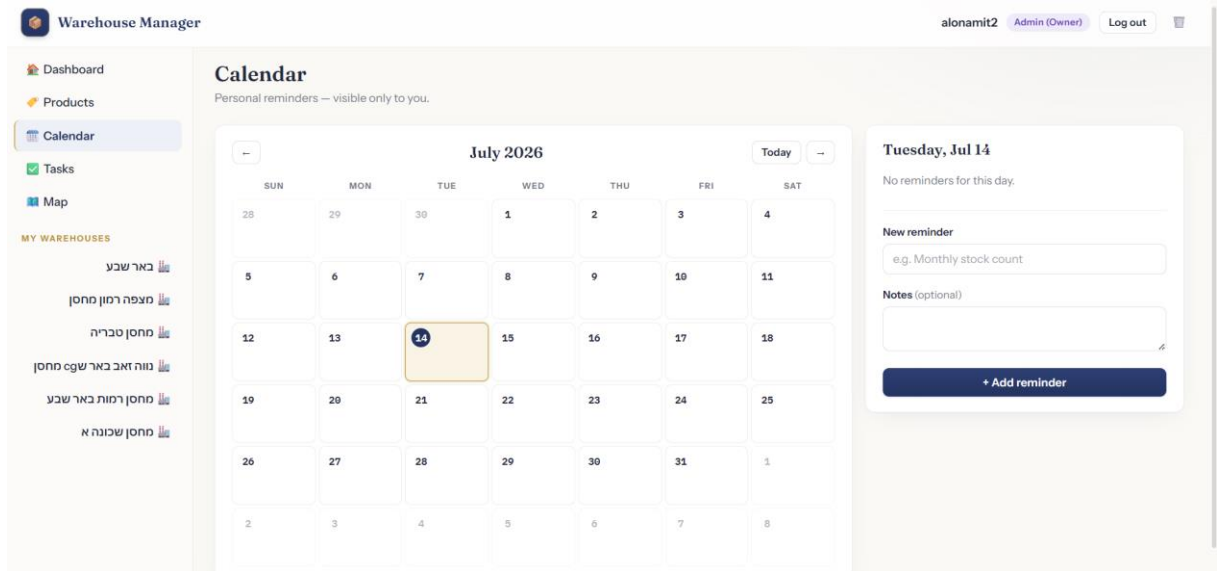
DATE	TEAM MEMBER	HOURS	NOTES
Tue, Jul 14	alon2	08:00-16:00	—

Schedule tab — assign a shift with day, hours and notes.

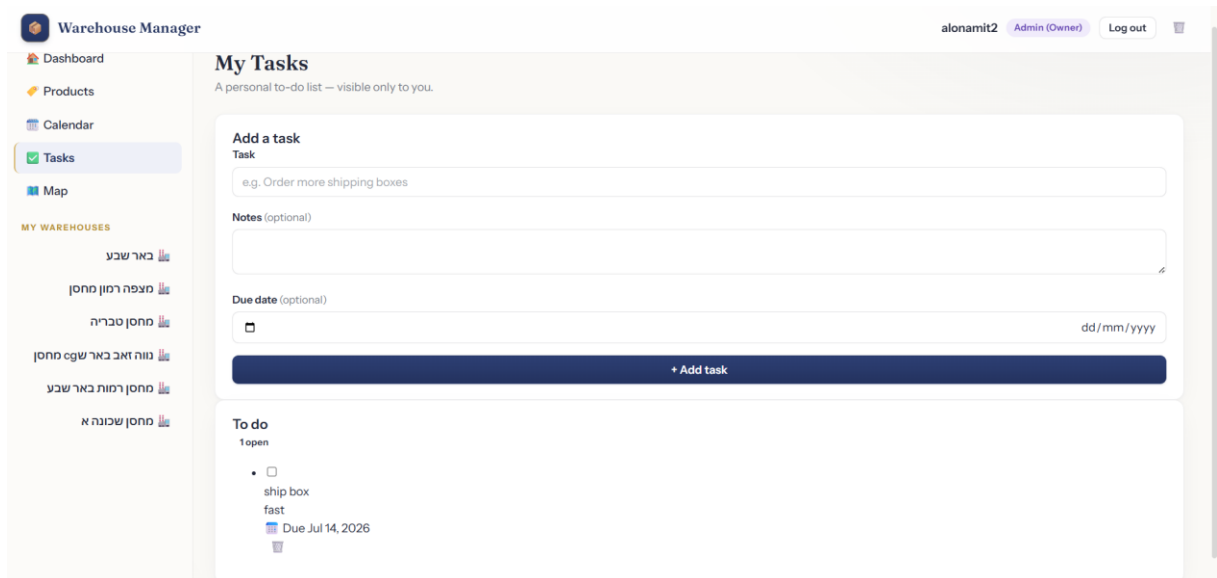
Personal productivity

- **Reminders** — pinned to a calendar date; today's reminders surface on the dashboard.

- **Tasks** — a personal to-do list with optional due dates, an overdue flag, and check-to-complete.



Calendar — private reminders pinned to a date.



My Tasks — personal to-do list with due dates and overdue flag.

Flagship — Smart route optimizer

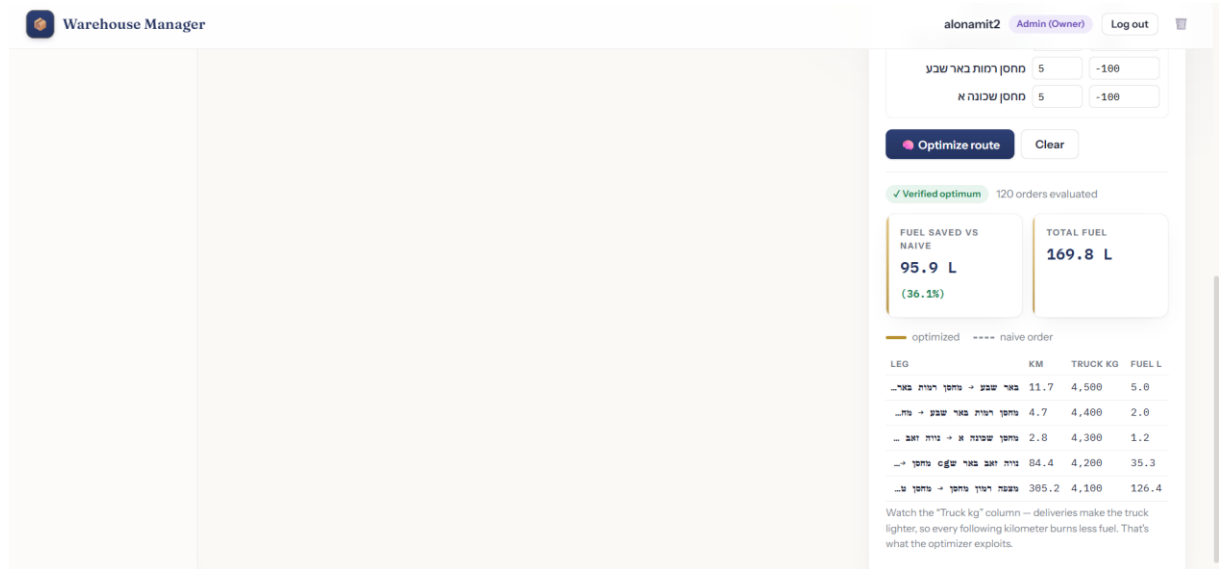
The optimizer models a truck that gets lighter as it delivers: fuel burn depends on the gross weight, and inventory urgency pushes low-stock stops earlier. It searches visiting orders (exhaustive for small sets, nearest-neighbour + 2-opt for larger ones) for the lowest fuel + urgency score.

FUEL SAVED VS NAIVE 95.9 L (36.1%)	TOTAL FUEL 169.8 L	SEARCH Verified optimum · 120 orders
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Route legs — watch the truck weight fall with every delivery, so each following kilometre burns less:

Leg	Distance (km)	Truck weight (kg)	Fuel (L)
Be'er Sheva → Ramot	11.7	4,500	5.0
Ramot → Shuna A	4.7	4,400	2.0
Shuna A → Neve Zeev	2.8	4,300	1.2
Neve Zeev → Mitzpe Ramon	84.4	4,200	35.3
Mitzpe Ramon → Tiberias	305.2	4,100	126.4

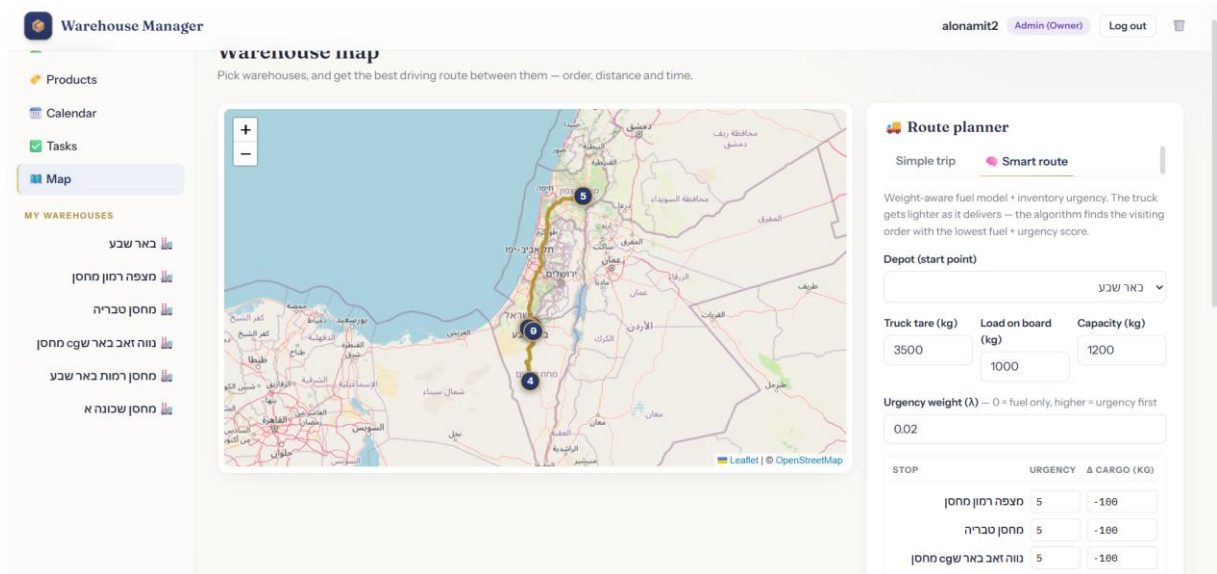
Truck: tare 3,500 kg · load 1,000 kg · capacity 1,200 kg · urgency weight $\lambda = 0.02$.



Smart route results — fuel saved, total fuel, and the per-leg weight breakdown.

Map & geocoding

- Warehouse addresses are geocoded to real coordinates on save.
- The planner draws the optimized path against the naive order on a Leaflet / OpenStreetMap map, reporting true driving distance and time — no paid map key required.



Warehouse map — the optimized route drawn between real warehouse locations.

Architecture & security

Frontend	Backend	Security
- Angular 22 (standalone, zoneless) - Signals + computed state - Functional HTTP interceptor - Lazy routes & guards	.NET 10, layered architecture - EF Core + SQL Server - Repository pattern, DTOs - Docker-ready, self-migrating	- JWT access token (1 h, stateless) - Refresh-token rotation in HttpOnly cookie - BCrypt, hashed tokens, rate limiting 72 tests — unit + integration

Testing

- Backend** — 44 xUnit tests: unit (route algorithm, transaction signs, auth, refresh rotation) + integration (WebApplicationFactory, in-memory DB, cookie flow).
- Frontend** — 28 vitest tests: auth service, HTTP interceptor auto-refresh, route guards, transaction sign logic.
- Total:** 72 tests, all green.

Warehouse Manager — inventory, logistics and a team in one clean full-stack app.